FitBuddy-AI Fitness Plan Generator Using Gemini Models

Government Arts and Science College-Manur-MSU142

TeamLeader:

Sandhiya P NM ID:F24D0CD882B6C7E9B9F44821FF74E70B

Team Members:Santhanamari G NM ID: C1A1D3256DDD23E87C0472B52EDBF20B

Saraswathi P NM ID: F7D8EF3617B74F0922A414AB1EDEF80E

Chinnathai S NM ID:6A593ADCC611910BA8240846346DA660

Muthumari M NM ID: 61F719397D001D3BE07494880A4743B7

Project Report

FitBuddy-AI Fitness Plan Generator Using Gemini Models

Therefore, the proposed FitBuddy system uses the Gemini AI model to generate an organized fitness plan based on user preferences. The system aims to simplify workout planning and provide useful exercise suggestions through a simple interface.

4\. OBJECTIVES

The main objectives of this project are:

* To develop an AI-based fitness plan generator.

* To collect basic fitness preferences from users.

* To integrate the Gemini generative AI model.

* To generate a weekly workout schedule based on user input.

* To provide exercise suggestions according to available equipment.

* To display the generated plan in a simple and understandable format.

* To include general recovery guidance and safety reminders.

5\. EXISTING SYSTEM

In the existing system, users generally depend on fitness websites, mobile applications, online videos, or pre-designed exercise schedules. These sources provide workout information, but users must often select a suitable plan and organize it according to their personal requirements.

Disadvantages of the Existing System

* Workout schedules may be generic.

* Searching multiple sources takes time.

* Beginners may find it difficult to select appropriate exercises.

* Available equipment and time may not match the selected routine.

* Users may have difficulty organizing a weekly workout schedule.

6\. PROPOSED SYSTEM

The proposed FitBuddy application uses Artificial Intelligence to simplify fitness planning. The user enters the required information, such as fitness goal, experience level, workout duration, available days, and equipment.

The application sends these details to the Gemini model through an API. The model generates a fitness plan based on the provided information. The application then displays the generated plan in an organized format.

Advantages of the Proposed System

* Personalized output based on selected preferences.

* Easy-to-use interface.

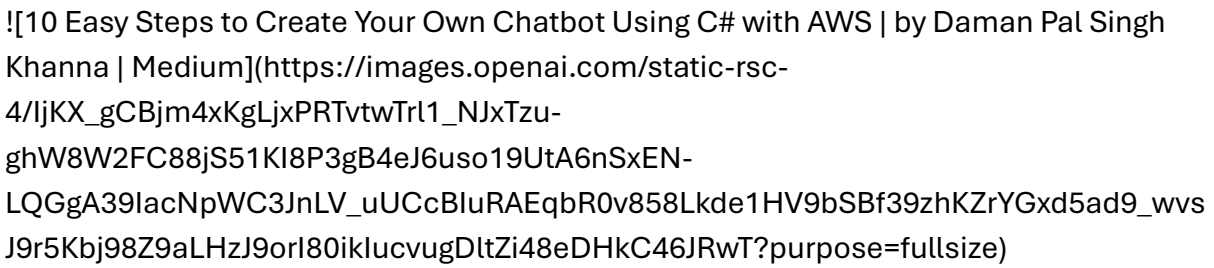
* Faster workout plan preparation.

* Structured weekly schedule.

* Support for different equipment preferences.

* Scope for future improvements.

7\. SYSTEM ARCHITECTURE



The system follows these steps:

* User: Enters personal fitness preferences.

* Frontend: Collects the input using a form.

* Backend: Validates the input and prepares a prompt.

* Gemini Model: Generates a fitness plan.

* Response Processing: Organizes the generated information.

* Output: Displays the plan to the user.

8\. METHODOLOGY

The project is developed through the following stages:

* Requirement analysis.

* User interface design.

* Input validation.

* Prompt preparation.

* Gemini API integration.

* Fitness plan generation.

* Output display.

* Testing and error handling.

9\. MODULE DESCRIPTION

User Input Module: Collects the user's fitness goal, experience level, workout duration, available days, and equipment.

Input Validation Module: Checks whether the required information has been entered correctly.

Prompt Generation Module: Converts the selected preferences into instructions for the Gemini model.

Gemini API Integration Module: Sends the prompt to the AI model and receives its response.

Fitness Plan Display Module: Presents the generated workout plan in a readable format.

Error Handling Module: Displays appropriate messages when input is missing or the API is unavailable.

10\. ALGORITHM

* Start the application.

* Display the fitness preference form.

* Collect user input.

* Validate the entered information.

* Prepare the prompt.

* Send the prompt to the Gemini model through the backend.

* Receive the generated fitness plan.

* Format the response.

* Display the plan and safety reminders.

* Stop.

11\. IMPLEMENTATION

The frontend can be developed using HTML, CSS, and JavaScript. A backend can be implemented using Python Flask, FastAPI, or Node.js. The Gemini API is used to generate the fitness plan.

The implementation begins with a form containing the required fields. When the user clicks the generate button, the application validates the input and sends it to the backend. The backend communicates with Gemini and returns the generated result.

The API key should be stored securely on the backend rather than directly in the frontend code.

12\. TESTING AND EXPECTED RESULTS

Testing verifies whether the application works correctly for different inputs.

Test case	Expected result
Valid user input	Fitness plan is generated
Missing fitness goal	Validation message appears
Bodyweight-only option	Prompt requests bodyweight exercises
API connection failure	Error message appears
Different workout duration	Prompt reflects selected duration

The expected outcome is an application that accepts user preferences and displays a structured fitness plan. Actual results and screenshots should be added after the application has been implemented and tested.

13\. LIMITATIONS

* An internet connection may be required.

* AI-generated suggestions may not always be suitable for every user.

* The system cannot observe exercise technique or assess physical condition.

* The quality of the result depends on the input and AI response.

* The application does not replace medical advice or professional fitness guidance.

14\. FUTURE ENHANCEMENTS

* User login and saved plans.

* Workout progress tracking.

* Workout reminders.

* Tamil and other language support.

* Calendar integration.

* Wearable device integration.

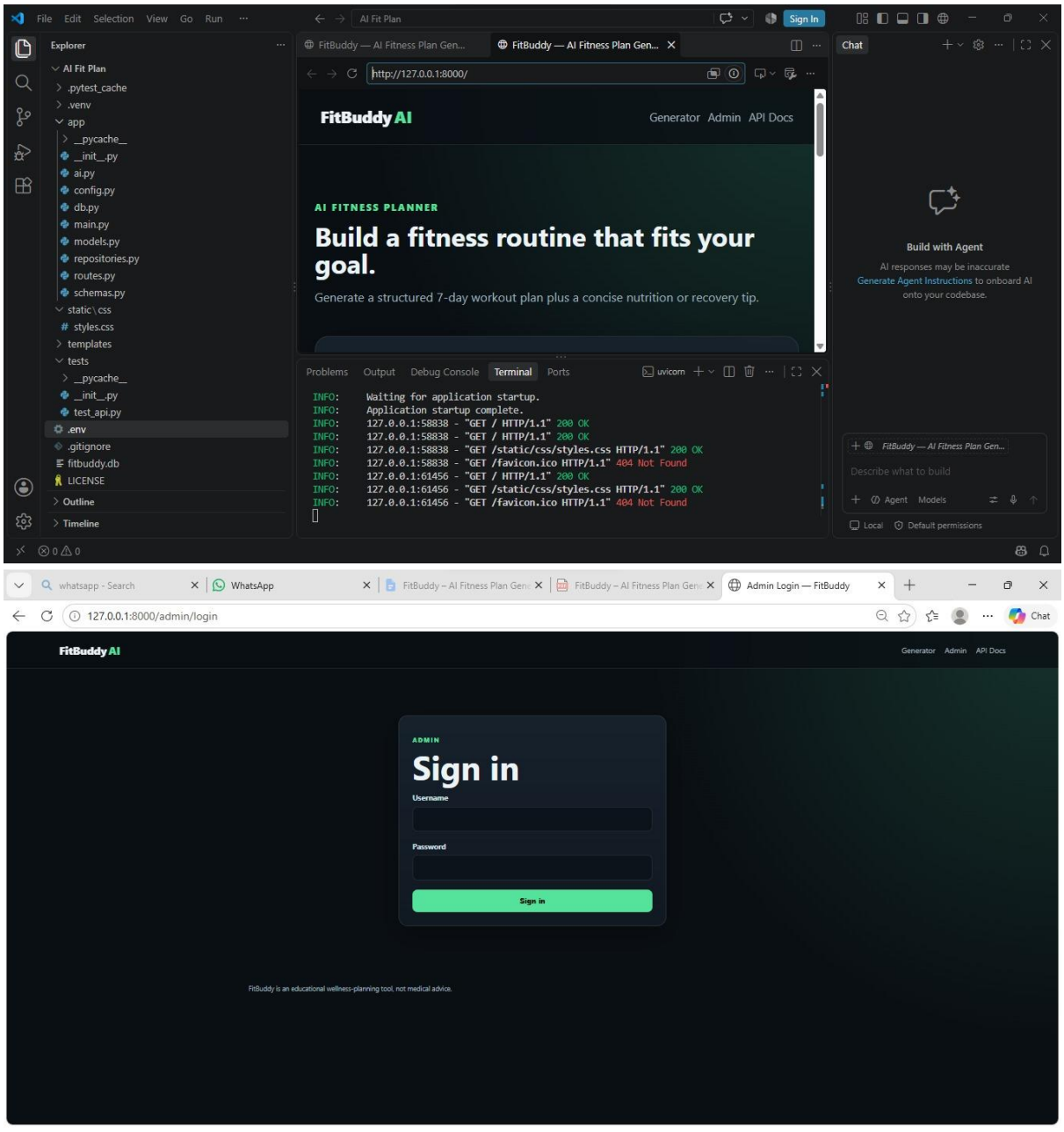
* Improved exercise substitutions and accessibility options.

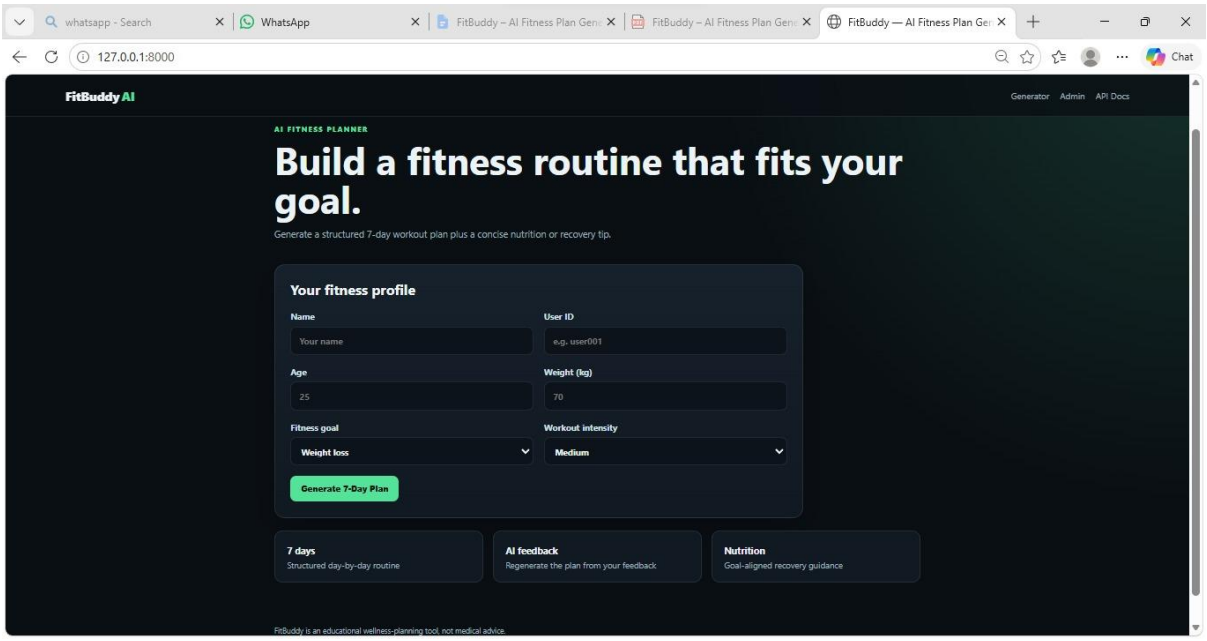
15\. CONCLUSION

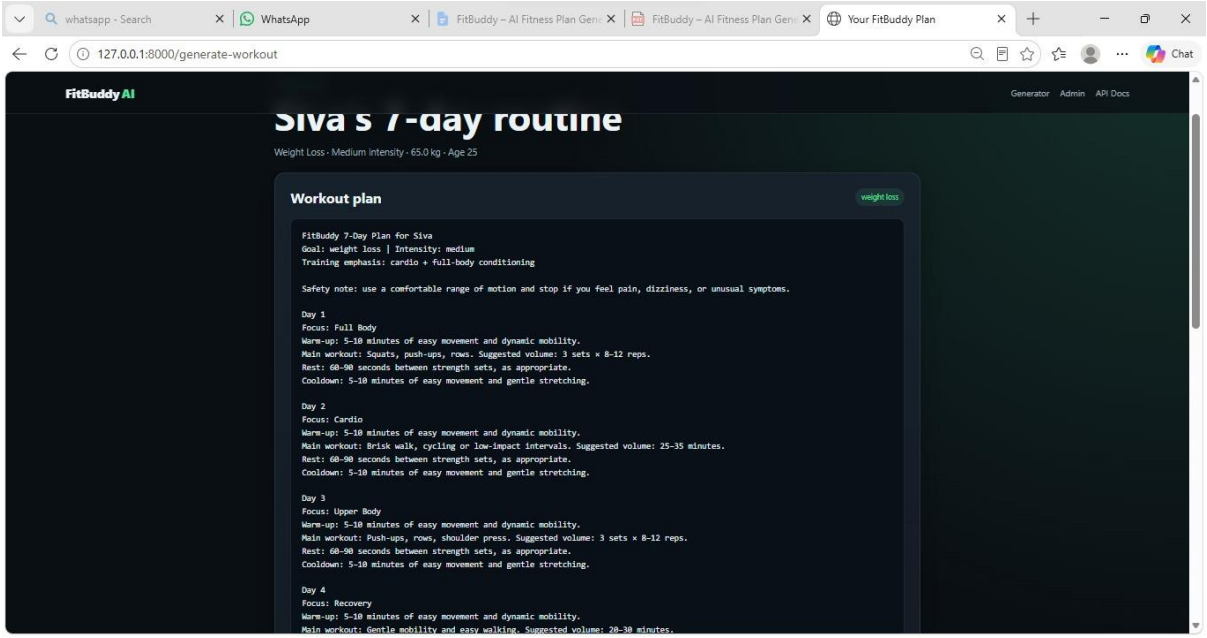
FitBuddy – AI Fitness Plan Generator Using Gemini Model is a proposed AI-based application designed to simplify the preparation of general fitness plans. The system collects user preferences and uses the Gemini generative AI model to create an organized workout schedule. The generated output can help users understand and arrange their fitness activities more conveniently.

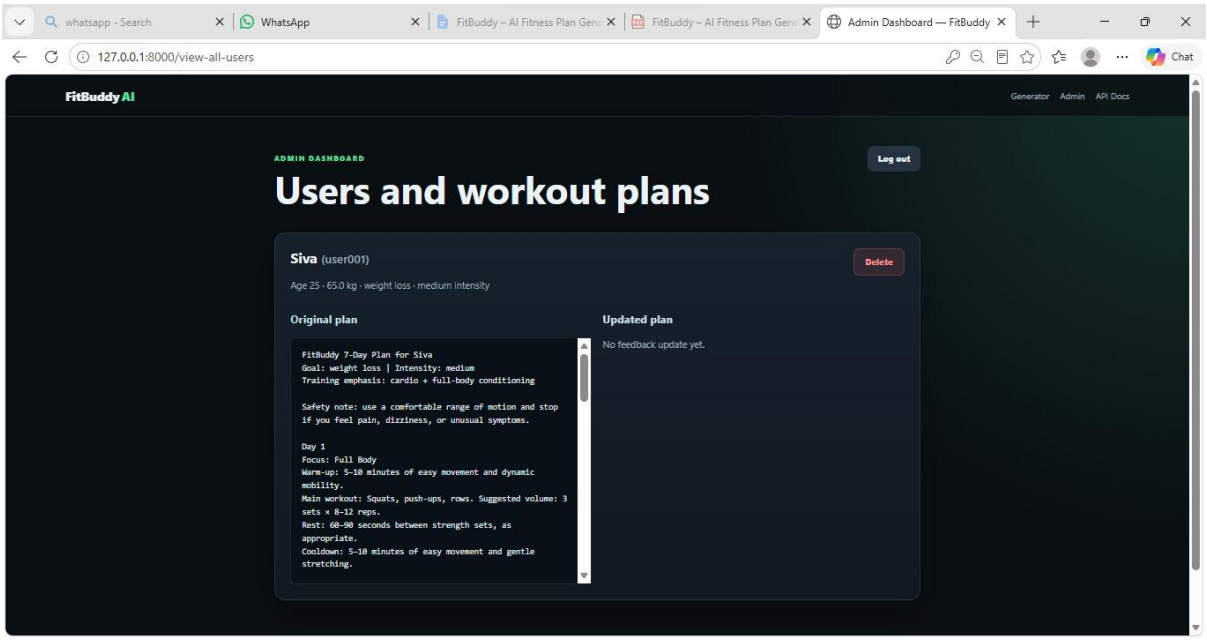
The project demonstrates the use of generative AI in a practical application. Secure API integration, suitable prompt design, input validation, testing, and safety reminders are important for developing a useful application

SCREEN OUTPUT :




whatsapp - Search
WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
FitBuddy — AI Fitness Plan Ger
127.0.0.1:8000
Chat
FitBuddy AI
Generator
Admin
API Docs
AI FITNESS PLANNER
Build a fitness routine that fits your goal.
Generate a structured 7-day workout plan plus a concise nutrition or recovery tip.
Your fitness profile
Name
Your name
User ID
e.g. user001
Age
25
Weight (kg)
70
Fitness goal
Weight loss
Workout intensity
Medium
Generate 7-Day Plan
7 days
Structured day-by-day routine
AI feedback
Regenerate the plan from your feedback
Nutrition
Goal-aligned recovery guidance
FitBuddy is an educational wellness-planning tool, not medical advice.


whatsapp - Search
WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
Your FitBuddy Plan
127.0.0.1:8000/generate-workout
Chat
FitBuddy AI
Generator
Admin
API Docs
Siva's 7-day routine
Weight Loss · Medium intensity · 65.0 kg · Age 25
Workout plan
weight loss
FitBuddy 7-Day Plan for Siva
Goal: weight loss | Intensity: medium
Training emphasis: cardio + full-body conditioning
Safety note: use a comfortable range of motion and stop if you feel pain, dizziness, or unusual symptoms.
Day 1
Focus: Full Body
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Squats, push-ups, rows. Suggested volume: 3 sets × 8–12 reps.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 2
Focus: Cardio
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Brisk walk, cycling or low-impact intervals. Suggested volume: 25–35 minutes.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 3
Focus: Upper Body
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Push-ups, rows, shoulder press. Suggested volume: 3 sets × 8–12 reps.
Rest: 60–90 seconds between strength sets, as appropriate.
Cooldown: 5–10 minutes of easy movement and gentle stretching.
Day 4
Focus: Recovery
Warm-up: 5–10 minutes of easy movement and dynamic mobility.
Main workout: Gentle mobility and easy walking. Suggested volume: 20–30 minutes.


whatsapp - Search
WhatsApp
FitBuddy – AI Fitness Plan Gene
FitBuddy – AI Fitness Plan Gene
Admin Dashboard — FitBuddy
127.0.0.1:8000/view-all-users
Chat
FitBuddy AI
Generator
Admin
API Docs
ADMIN DASHBOARD
Log out
Users and workout plans
Siva (user001)
Delete
Age 25 · 65.0 kg · weight loss · medium intensity
Original plan
Updated plan
No feedback update yet.
FitBuddy 7-Day Plan for Siva
Goal: weight loss | Intensity: medium
Training emphasis: cardio + full-body conditioning
Safety note: use a comfortable range of motion and stop if you feel pain, dizziness, or unusual symptoms.
Day 1
Focus: Full Body
Warm-up: 5-10 minutes of easy movement and dynamic mobility.
Main workout: Squats, push-ups, rows. Suggested volume: 3 sets x 8-12 reps.
Rest: 60-90 seconds between strength sets, as appropriate.
Cooldown: 5-10 minutes of easy movement and gentle stretching.